Level 2 · about 14 h · Python 3.8+

Project: A number-guess game
Examples run on CPython 3.12.3. **You will be able to:**

- Write if, elif and else branches with comparison and boolean operators
- Loop with for and range, and with while
- Use break and continue and say when each is clearer
- Trace a small program by hand, line by line

Decisions and Loops

Decisions with if

Python runs the first branch whose test is true. Indentation shows which lines belong to the branch.

```
score = 72
if score >= 90:
    grade = "A"
elif score >= 70:
    grade = "C or better"
else:
    grade = "keep practising"
print(grade)
```

```
Output (CPython 3.12.3):
C or better
```

Comparing and combining

Comparisons give booleans. Combine them with and, `or`, `not`.

```
age = 15
has_ticket = True
print(age >= 13 and has_ticket)
print(not has_ticket)
```

```
Output (CPython 3.12.3):
True
False
```

Loops with for and range

`range(n)` counts from 0 up to n - 1.

```
for i in range(3):
    print("lap", i)
print(sum(range(1, 6)))
```

```
Output (CPython 3.12.3):
lap 0
lap 1
lap 2
15
```

Loops with while

A while loop repeats as long as its test is true. Change something inside, or it never ends.

```
n = 3
while n > 0:
    print(n)
    n -= 1
print("go")
```

```
Output (CPython 3.12.3):
3
2
1
go
```

break and continue

`break` leaves the loop. `continue` skips to the next pass.

```
for i in range(10):
    if i % 2 == 0:
        continue
    if i > 7:
        break
    print(i)
```

```
Output (CPython 3.12.3):
1
3
5
7
```

Tracing by hand

Write each name and its value after every line. Compare your table with the output.

```
total = 0
for i in range(1, 4):
    total += i
    print(i, total)
```

```
Output (CPython 3.12.3):
1 1
2 3
3 6
```

Check yourself

1. Which branch runs when `score` is 70?
2. How many times does `for i in range(3)` run its body?
3. What does `continue` do?

Project: Number-guess game

The computer picks a number from 1 to 20 (use a fixed number first). Loop until the guess is right, print "higher" or "lower", and stop after 6 tries.

Sources

- Python tutorial: More control flow tools
- PEP 8 Style Guide for Python Code
- Python docs: Truth value testing

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